

Heart rate variability metrics for fine-grained stress level assessment



Tânia Pereira^{a,*}, Pedro R. Almeida^b, João P.S. Cunha^c, Ana Aguiar^a

^aTelecommunications Institute, University of Porto, Portugal

^bSchool of Criminology, University of Porto, Portugal

^cINESC TEC and Faculty of Engineering, University of Porto, Portugal

ARTICLE INFO

Article history:

Received 31 July 2016

Revised 27 March 2017

Accepted 23 June 2017

Keywords:

Heart rate variability

Stress assessment

Wearable technology

Trier social stress test

ABSTRACT

Background and Objectives: In spite of the existence of a multitude of techniques that allow the estimation of stress from physiological indexes, its fine-grained assessment is still a challenge for biomedical engineering. The short-term assessment of stress condition overcomes the limits to stress characterization with long blocks of time and allows to evaluate the behaviour change in real-world settings and also the stress level dynamics. The aim of the present study was to evaluate time and frequency domain and nonlinear heart rate variability (HRV) metrics for stress level assessment using a short-time window. **Methods:** The electrocardiogram (ECG) signal from 14 volunteers was monitored using the Vital JacketTM while they performed the Trier Social Stress Test (TSST) which is a standardized stress-inducing protocol. Window lengths from 220 s to 50 s for HRV analysis were tested in order to evaluate which metrics could be used to monitor stress levels in an almost continuous way. **Results:** A sub-set of HRV metrics (AVNN, rMSSD, SDNN and pNN20) showed consistent differences between stress and non-stress phases, and showed to be reliable parameters for the assessment of stress levels in short-term analysis. **Conclusions:** The AVNN metric, using 50 s of window length analysis, showed that it is the most reliable metric to recognize stress level across the four phases of TSST and allows a fine-grained analysis of stress effect as an index of psychological stress and provides an insight into the reaction of the autonomic nervous system to stress.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

The impact of the mental stress in the health condition is currently well recognized [1–3]. The effect of the stress on the human body capability to regulate inflammation can promote the development and progression of diseases. Stress was identified as a risk factor in the heart diseases, diabetes, asthma, upper respiratory infections and depression [1,2,4,5]. For these reasons, it is extremely important to have instrumental solutions that allow the stressful events assessment and monitoring in order to measure, quantify and correlate with health impact [4,6–8].

In the last two decades several efforts have been made to apply sensors and novel technologies for stress detection and assessment [9–11]. Conventionally, stress levels were estimated by assessing hormonal responses and using questionnaires [12]. However, these methods are not useful in a real-world context, also referred to as ecological settings [13]. A method for stress assessment in real context should be minimally obtrusive in order to allow the sub-

jects to perform their tasks with complete freedom, and it should allow to monitor stress levels in an almost continuous way [4].

Several non-invasive measurements have been considered in the literature for ecological studies. These include the electrocardiogram (ECG), electroencephalogram (EEG), body temperature, electromyogram, galvanic skin response and blood volume pulse [10]. The metrics from ECG have been used as a marker of the autonomic modulation of the heart, and recent developments in wearable technology made it possible to monitor this physiological signal in non-invasive and non-obtrusive ways. Heart rate variability is an established, non-invasive and quantitative method of assessing cardiac autonomic activity [7,8,14].

Although HRV was firstly used in recordings from 5 min to several hours in length (long-term recordings), currently it is recognized that HRV from records of less than 2 min duration (short-term recordings) can be used to accurately assess cardiac autonomic activity [14]. The use of long or short-term recordings depends on the study goal [15,16]. The short-term (≤ 5 min) analysis represents an advantage for stress studies due to the rapid physiological response time. Table 1 summarizes recent works that evaluated the differences in HRV under stress vs baseline using short time windows. The first studies that used short window length

* Corresponding author.

E-mail address: taniapereira10@gmail.com (T. Pereira).

Table 1
Summary of HRV studies using short-time windows for stress assessment.

Subjects (n) [Reference]	Age (years)	Procedure	HRV	Windows length
738 [17]	–	3 stressors (virtual reality car driving - video game challenge - social competence interview)	rMSSD and HF	30 s
19 [18]	17–27	Stressor (flight simulator)	mid- and high- frequency bands	30 s
24 [19]	26 ± 4.6	Rest and Stressor (stroop test)	AVNN, rMSSD, pNN50, HF, LF/HF, LFnu and HFnu	50 s
48 health 48 major depression (MD) [8]	Health: 21 ± 2; MD: 23 ± 2	Neutral and emotionally audiovisual stimuli	LF, HF, LF/HF, ApEn, SampEn	90 s
25 [21]	21.36 ± 2.97	Baseline and Stressor (TSST)	HR, SDNN, rMSSD, pNN50, LF, LFnu, HF, LF/HF	3–5 min
92 [22]	20–25	Baseline and Stressor (university examination)	AVNN, SDNN, rMSSD, pNN50, Mo, AMo, LF, HF, LF/HF	5 min
42 [23]	–	Baseline and Stressor (university examination)	SD1, SD2, En, D ₂ , α ₁ , α ₂ , REC, DET, ShEn	5 min
18 [24]	18.7 ± 0.69	Baseline and Stressor (university examination)	HR, SDNN, pNN50, RMSSD, VLF; LF, HF, LFnu, HFnu and LF/HF	5 min

ApEn, approximate entropy; Mo, mode (geometrical metric); AMo, mode amplitude (geometrical metric); REC, recurrence rate (nonlinear metric); DET, determinism (non-linear metric); ShEn, Shannon entropy (nonlinear metric).

spurred the exploitation of the idea that HRV provides insight on mental effort intermediate variations, however, only a low number of metrics were tested (rMSSD and HF) [17,18]. The work that tried to evaluate how short ECG recordings could be reliably used for mental stress monitoring concluded that it is possible within 50 s of window length [19]. Nevertheless, the referred work studied a small group of metrics and tested only two situations, the baseline stage and stroop color word test [19]. Time domain metrics showed low repeatability for ten-seconds segment analysis [14], therefore, larger windows should be used. The frequency domain metrics limit the segment length for analysis, for that reason small segments do not allow the determination of lower frequency bands [17,18,20]. The heart rate (HR) series length should be defined by the balance between the time window length that allows a fine-grained analysis of stress level and the resolution of the spectral estimation. The longer the recording time, the better spectral resolution because it would be based on more data points, however, long analysis windows preclude fine-grained stress monitoring.

Several studies have shown an association between HRV and stress, and use HRV as a psychophysiological index of mental stress. A significantly lower mean RR interval (time between R waves in QRS) i.e., higher HR, was measured during a mental task than in a control condition, while pNN50 was significantly higher in the control condition than within a mental task [22,25]. For the SDNN and rMSSD metrics a significant decrease with stress condition was verified [22]. The mean LF during cognitive stress sessions was significantly lower than during the rest sessions [22,26]. The nonlinear metrics were used for stress assessment and showed that they could be effective in stress detection [23].

The Trier Social Stress Test (TSST) is an effective laboratory protocol for inducing mental stress by exposing individuals to situations that constitute social evaluative threat [27]. Heart rate monitoring with a wearable device was used for stress assessment and the obtained HR was compared with cortisol levels during TSST [27]. The study concluded that assessment of the psychobiological stress response during TSST is a better measure as compared to post-TSST values [27]. This important conclusion is a motivation to assess the stress levels with fine-grained analysis that allows stress condition monitoring on an almost continuous way.

Guided by the previous studies and using novel wearable technology, the current work uses the TSST method to examine how four contextual conditions (silent, reading, public presentation and counting arithmetic task) influence the heart rate variability [12]. Since standard laboratory stressors do not always engage subject's affective response, social interaction stressors such as public speak-

ing tasks are applied to provide a more appropriate social context [28]. The TSST tasks involve active cognitive processing and responses, and are generally associated with varied degrees of parasympathetic withdrawal and sympathetic activation [29]. The TSST was applied to 14 volunteer participants who were equipped with the Vital Jacket™, which is a comfortable wearable biomedical equipment that records high quality ECG.

This work represents a global evaluation of HRV metrics on an healthy and young sample in a lab context that emulates the real-world. Time and frequency domain and nonlinear HRV metrics were used to assess the effect of mental stress, analysing which are reliable as stress indicators in a short-term analysis. The main objective of this work was to determine which metrics could be used for a fine-grained stress levels assessment. The changes in HR occur very rapidly and are caused by shifts in autonomic nervous system [30]. The sympathetic nervous system regulation has an initial response delay of up to 5 s, followed by a progressive change and a maximum response after 20 to 30 s [31]. In this work an attempt was made to assess these changes during the TSST proceeding. The window size chosen for the HRV analysis try to approximate time length with the physiological response of mental stress and monitor the changes along the TSST activities. The 50 s window size provides the best approximation of the physiological response, and short-term analysis reflect the dynamic response of the cardiovascular control system to a physiological perturbation induced by the stress and for this reason the HRV metrics were studied for different windows sizes in order to determine which metrics could discriminate between stress levels with small windows of analysis.

2. Materials and methods

The TSST was used to induce stress in a lab context that emulates the real-life condition. The State-Trait Anxiety Inventory (STAI) is a self-reported assessment of stress that was performed at non-stress phase and after the stress period [32]. The ECG signal was recorded with the Vital Jacket™ system in order to extract HRV and to obtain a fine-grained stress assessment.

2.1. Study population

Fourteen participants (five female) aged between 20 to 26 years participated in the study. No participant had an history of cardiac or mental disease. Only one subject was smoker, none of the subjects were regular consumer of alcohol and 4 practiced sport. Participants were volunteers and their physiological signals were

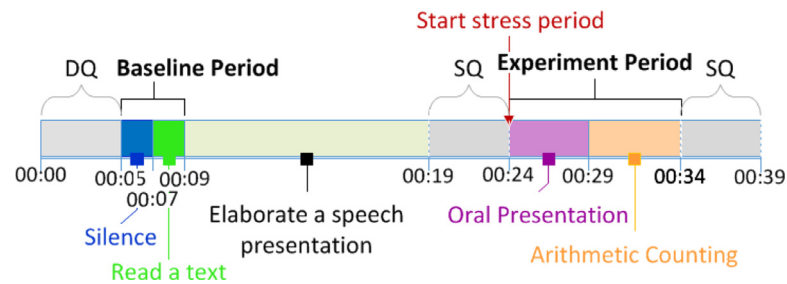


Fig. 1. Timeline in minutes of TSST phases (DQ: demographic questionnaire, SQ: STAI questionnaire).

monitored, as defined by the purpose of the study. All participants signed an informed consent, and the data collection was approved by the Portuguese Commission for Data Privacy.

Later, signals acquired from 2 subjects were excluded from the study due to reliable criteria of analysis based in the ratio NN/RR [33]. NN/RR is the fraction of total RR intervals that are classified as normal-to-normal (NN) intervals and included in the determination of HRV statistics. These two subjects were excluded because the data was considered unreliable due to having an NN/RR ratio below 70%, i.e. less than 70% of the RR intervals were considered NN.

2.2. Experimental procedures

The TSST consists mainly of a baseline period and an experiment period described in Fig. 1. Before knowing the complete task that they would perform, the participants filled the demographic questionnaires and were asked to record a baseline, without the presence of a stressor. They were instructed to keep silent for two minutes keeping their mind clean of any stressful thought. After that, they read a newspaper text with no emotional connotation. The physiological data was recorded for the baseline.

They were then informed of the task they would perform in the experiment phase. They were given ten minutes to elaborate a job interview speech that should last five minutes, and were then left alone in the room with pen and paper. After ten minutes, two unknown individuals entered the room, the participants were told they were jury for the job position, and the video camera present in the room would be used to record their performance for later evaluation. The video camera is part of the TSST protocol, but it was not recording. Participants had then five minutes to introduce themselves as job applicants. If they finished before the five minute mark, the jury then said “You still have time, please continue”, if they finished again the jury would be quiet for twenty seconds and then ask a series of prepared questions in order to fill the five minutes. Finally, participants were to sequentially subtract 13 from a number, starting at 1022. Upon every failure the participant had to restart at 1022. This process lasted for 5 min. During all the tasks, the physiological data was recorded. When the test was over, the participants filled out the STAI questionnaires. At the end of the procedure, the participants were debriefed and a full explanation of the test was given. Fig. 1 shows the timeline overview of the test, identifying the sequence of tasks and the timings of each phase.

The set up conditions were replicated equally for all participants, both in terms of room disposal and previous knowledge of test conditions. At the end of each test, two recordings were obtained. The baseline from the period of no stress activation during the silence and read task that lasted typically between four and five minutes, and the recording of the TSST itself, henceforward called event recording, which lasts between ten and twelve minutes and represents the period of stress activation.

2.3. Psychological measurements

Psychological stress was assessed using the Portuguese version of the STAI [34]. The instrument has been successfully used to evaluate stress/no stress conditions [32]. It consists of 20 questions, and the scale uses a 4-point Likert type response scale anchored at 1 = Not at all and 4 = Very much. Good psychometric properties (reliability and fit indicators) have been reported for this scale [34].

2.4. Physiological measurement system

During the TSST procedure, the ECG was recorded by the Vital Jacket™ (VJ™) using 1 lead deviation for ambulatory ECG long term use. The VJ™ is a wearable vital signal monitoring device that was designed in form of a t-shirt with embedded microelectronics that allows to record 1 lead clinical quality ECG with sampling frequency rate of 500 Hz and 3-axis actigraphy [35,36]. The VJ™ is a certified medical device with the EU CE 1011 Medical Device certification clearance and is compliant with the EU directive 42/93/CE and certified to comply with the standards ISO9001 and ISO13485 [6]. The VJ™ system allows recording the high quality ECG, from which the R-R intervals are extracted, that correspond the time between peaks in the electrocardiogram. An embedded box processes and stores the signal on a SD memory card and transmits it directly to another device using a bluetooth connection.

2.5. Heart rate variability metrics

The aim of this study was to find HRV parameters associated with stress calculated on short time windows. The HRV was analysed in the time-frequency domain and with nonlinear metrics. Time domain measures of heart rate variability include various statistics that are calculated from the time intervals between heart beats (RR) [15]: AVNN represents the average value of NN; SDNN is a standard deviation of NN intervals; rMSSD is the root mean square successive difference that evaluates the differences between successive NN intervals, and thus describes short-term variability. The parameter pNN20 is the ratio of the consecutive NN intervals differing by more than 20 ms divided by the total number of NN intervals, and pNN50 is the ratio of adjacent NN intervals differing by more than 50 ms to all NN intervals [37].

Spectral measures from low frequency bands (LF: 0.04 and 0.15 Hz), high frequency bands (HF: 0.15 and 0.4 Hz) and the ratio of LF/HF are frequently used [25]. The normalized spectral index was determined and was defined as $LF_{nu} = LF / (LF + HF)$ [38].

Additionally, the following non-linear methods were estimated: Sample Entropy (SampEn) which quantifies irregularity of heart-beat fluctuations [39]; Detrended Fluctuation Analysis (DFA) which determine the overall root-mean-square fluctuations of integrated and detrended RR interval; Poincaré plot analysis which is a common method of adjacent interbeat interval analysis [40]. For the entropy determination, an embedding dimension $m = 3$ was used

and $r = 0.2$ was adopted as a threshold to identify irregularities [41,42]. The two slope parameters from the DFA log-log plot, α_1 (over the range of $4 \leq n \leq 16$ heartbeats) and α_2 (over the range of $16 \leq n$ heartbeats) represent short and long term fluctuations, respectively [43,44]. The metric α_2 was not used since a small window length will not contain enough beats for this parameter determination [45]. The Poincaré plot is formed by a scatter plot of RR_{i+1} Vs RR_i that compares the relationship between adjacent RR interval times. The minor axis SD1 (standard deviation of the short-term RR-interval variability), the major axis SD2 (the standard deviation of the long-term RR-interval variability), and the SD1/SD2 ratio were determined [46]. The SD1 and SD2 parameters measure the variability of the time series data.

Ectopic beats introduce a bias into HR variability results and represent a significant problem in the interpretation of these results [47]. The pre-processing of data comprises outliers removal, which excluded any RR interval less than 0.3 s or greater than 2.0 s. A percentage filter was applied in order to remove the ectopic beats that would replace every value that was deflecting more than 20% of the previous interval of RRs [15]. The percentage filter uses a window of 40 intervals (20 intervals on either side of the central point), the average over the window is calculated excluding the central interval. The abnormal values are replaced by the linear interpolation [48].

The algorithms provided by Physionet and the HRV Toolkit were used to process the RR sequences from the ECG signals and to determine the time and frequency domain HRV metrics [33]. The Physionet toolkit uses the Lomb Periodogram for spectral analysis. Power spectral density (PSD) estimation using FFT method requires resampling, but the Lomb-Scargle algorithm does not require it, producing robust PSD estimates even in the presence of noise and ectopy [49,50]. The nonlinear metrics were determined using the HRVAS: HRV Analysis Software tool developed using MATLAB™ [51]. Heart rate signals are dependent on each individuals physiological level. Hence, to eliminate the intra-individual factor the HRV metrics were standardized [52].

2.6. Windows length analysis

A short-time window stress metric was chosen to create a finer-grained picture of the autonomic nervous system reaction to different stressors encountered throughout the TSST phases. Different lengths of HR series were analyzed in order to evaluate which are the best metrics to distinguish the stress condition in short-time windows. HRV metrics were determined for window lengths of 220 s, 150 s, 100 s and 50 s. In this study, a decreasing sequence of window length was used in order to evaluate which metrics maintain the capability to distinguish the stress levels, for this reason we started with higher lengths and tested some intermediates values before the 50 s (the value for the physiological time response for the stress induced during the TSST). The windows used in this study were not sliding windows, and there was no overlap between them.

For each TSST phase, the collection of windows for analysis was defined using the first window located at the beginning of the period and using the subsequent time intervals until the end of the phase duration. These values were selected in order to use the maximum length of the shortest TSST phase. The baseline period comprises 240 s however some subjects take slightly less time and the window length of 220 s was chosen to ensure the same length window analysis for all samples. Intermediate dimensions were tested in order to analyze which metrics allow to distinguish between TSST phases, until the 50 s limit. The shortest signal length considered was 50 s that ensures the HRV metrics determination reliability and accuracy [53], and the best approximation of the physiological time response [31].

The minimum value of the frequency in the analysis is determined by the inverse of HR by the length of window analysis. The window of 50 s is long enough to offer an accurate estimation of the LF index [20]. The RR intervals were calculated directly from the ECG signal. The new sampling rate is defined by the RR intervals [20]. We can suppose that the RR signal repeated itself (periodic) and then we get that the samples of RR only give a unique representation of a signal if we assume this is band-limited to the frequency inverse of the mean heart period (RR interval) [20,54]. The maximum frequency with physiological meaning which can be analyzed is defined as half of the mean HR in the analyzed interval. The normal upper frequency limit of the highest band for HRV analysis is 0.4 Hz (upper limit of HF total power parameters) [55]. In this case, the band-limited for the analysis is determined to the frequency inverse of the mean heart period (RR interval) [54], and ensures the limits of HF parameter. Task force's states that HRV metrics from segments of different duration are not comparable [15]. For this reason, the results of different TSST phases were analyzed and compared among same length window.

To avoid the complications involved in assessing long term HRV introduced through the highly non-stationary nature of the RR tachogram, only short term periods are analysed (220 s to the 50 s windows length), where the data can be considered to be stationary or quasi-stationary. Changes in stationarity of RR-intervals are often joined by transient sections of heart rate overshoot and an accompanying increased probability of artifact on the ECG (and hence missing data) [54,56]. In this work the windows length selected never include the baseline and the experiment phase in the same interval of analysis.

2.7. Statistical analysis

Across all subjects, HRV parameters were calculated for each selected time period window and those metrics were expressed as mean and standard deviation. In order to evaluate whether there is statistical evidence that the means of the HRV metrics differ over TSST, a one-way analysis of variance (ANOVA) was used [57,58]. For the HRV metrics that ANOVA led to a conclusion that their means differ between the TSST phases, the Tukey multiple comparison test was used to determine which HRV metrics had significant differences between the four TSST phases, i.e. silence-reading, silence-presentation, silence-counting, reading-presentation, reading-counting and presentation-counting. The nonparametric Wilcoxon Signed Rank test was applied to the STAI scores in order to investigate the difference between TSST phases. A value of $p < 0.05$ was considered statistically significant. The statistical software R was used for data processing and analysis.

3. Results

3.1. Psychological stress

STAI scores showed that participants had higher levels of stress during the experiment (42.13 ± 14.65) compared to the baseline condition (33.75 ± 8.78), and Wilcoxon test confirms the significant difference in the two stages ($Z = 2.366$, $p = 0.018$, two-sided). STAI is a subjective assessment tool of stress condition that indicates if the stress increases during the TSST but it cannot compare the increase rate between subjects and for this reason the correlation with HRV metrics does not show statistical significant results but shows the relationship (positive or negative correlation) between STAI evaluation and HRV metrics. All the metrics were negatively correlated with STAI score [59], excepted the LF/HF, LF_{nu} and α_1 which show a positive correlation. HRV metrics are normalized for each subject, however STAI only gives the information whether

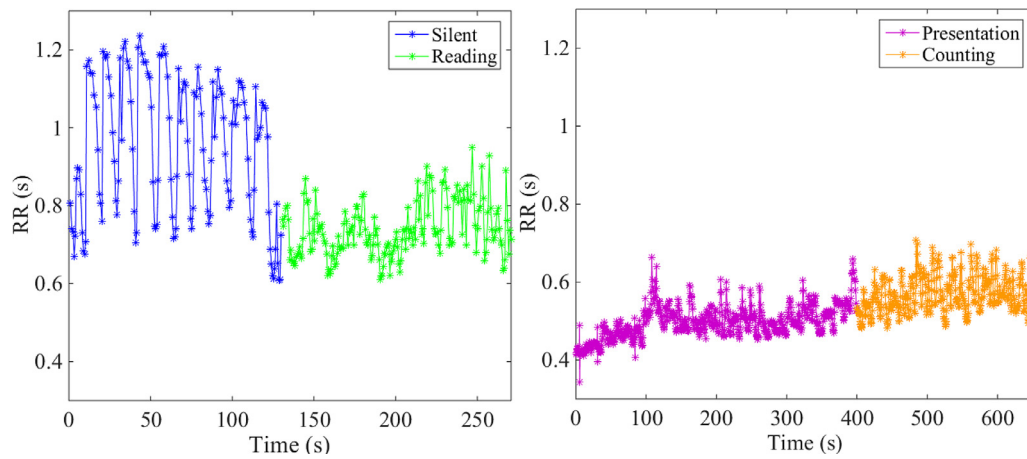


Fig. 2. The RR tachogram plot of one subject during the baseline and experiment period.

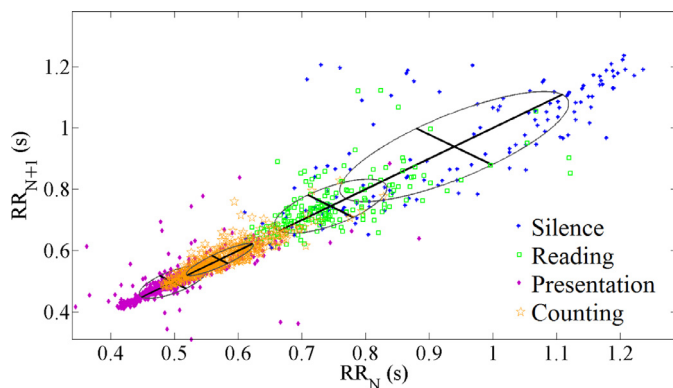


Fig. 3. Poincaré plot of RR intervals for one subject with standard descriptors SD1 and SD2 for each phase of TSST.

the stress condition for each subject increases or not. The subjects were asked to rate their anxiety state during the TSST but this represents a subjective perception. For this reason, the relationship was carefully considered and the results from the STAI were just used to confirm the increase of the stress during the experiment phase [60].

3.2. Physiological signal

Fig. 2 shows the RR intervals for one subject during the TSST procedures. The RR tachogram plot allows the identification of the changes on HRV across the different TSST phases through visual inspection. Poincaré plot analysis is another visual inspection of RRs (Fig. 3). It is valuable due to its ability to display nonlinear aspects of the data sequence because each phase of TSST presents a different shape of the data dispersion. The SD1 and SD2 parameters for the TSST phases represented in the Fig. 3 reveal the differences.

The methodology proposed in this work tested several window lengths to evaluate the metrics that are best suited for mental/emotional stress detection in fine-grained analysis. The windows size of 220 s comprises the silence and reading period, which is referred to as baseline phase, and the experiment phase with presentation and counting stages. Fig. 4 shows the results for AVNN and LF metrics and the differences between the metrics for the TSST stages are clearly visible. For this window length of 220 s all the metrics in analysis (time, frequency and nonlinear) show consistent differences between baseline and experiment phases, with p -values smaller than 0.0001. With the decrease of the window size in analysis, 220 s, 150 s, 100 s and 50 s, some metrics

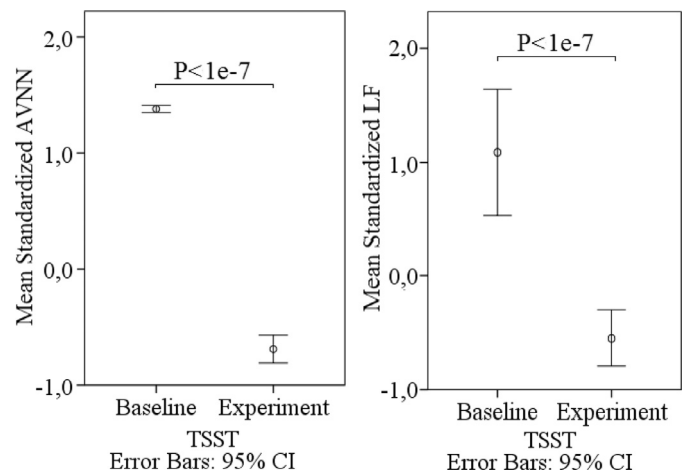


Fig. 4. AVNN and LF parameters for 220 s of length window of analysis.

lose the significant difference between the stress phase. These intermediates sizes were used in a progressive test in order to evaluate which metrics maintain the capability to distinguish the stress condition in the short-term analyses, revealing significant differences.

Table 2 summarize the average values obtained by the HRV metrics for all windows in each activity and across all subjects. The windows length of 50 s was used for the TSST phases: silence, reading, presentation and counting. The reference values of HRV from other studies were represented, and in these cases the values refer to healthy subjects in the rest condition.

The HRV values obtained in the TSST are in agreement with reference values, however, the slight differences could be explained by the age range of samples that influence the HRV [43] and in this case it was very restricted. A constant decline in autonomic nervous system activity due to a significant decrease in cardiac parasympathetic activity was verified [43,61]. Increasing age is associated with a reduction in overall HRV [43,61], which is in accordance with the data showed in the Table 2, where the reference values for some metrics are closer to the stress situation than the non-stress, revealing the probable effect of the age.

The segment analysis of 50 s windows allows a fine-grained of stress assessment, however, only some metrics show significant differences between TSST phases. The repeated-measures ANOVA results showed which HRV metrics have significant differences between the stress conditions (see Table 2). The p -values presented

Table 2
Values of HRV measurements determined for 50 s window length during TSST phases and reference values. Significance levels obtained by analysis of ANOVA on the TSST phases.

	Reference Values	Silence	Reading	Presentation	Counting	<i>p</i> -value
AVNN (ms)	926 ± 90 [62]	783.13 ± 129.22	701.58 ± 100.71	572.83 ± 113.43	606.42 ± 108.66	***
SDNN (ms)	50 ± 16 [62]	49.51 ± 24.63	45.94 ± 19.24	32.70 ± 16.90	38.05 ± 18.46	***
rMSSD (ms)	42 ± 15 [62]	42.75 ± 25.68	35.37 ± 21.41	21.34 ± 12.78	24.13 ± 13.51	***
pNN20 (%)	52.3 ± 12.1 [59]	52.12 ± 29.61	40.84 ± 21.90	24.16 ± 19.06	29.52 ± 18.20	***
pNN50 (%)	23.6 ± 11.1 [59]	29.67 ± 32.39	12.16 ± 12.11	5.39 ± 8.00	5.63 ± 7.56	***
LF (ms ²)	1847 ± 1067 [59]	1208.24 ± 1291.87	1134.89 ± 891.36	688.52 ± 805.07	931.13 ± 969.53	***
HF (ms ²)	975 ± 676 [59]	1422.71 ± 1669.12	826.38 ± 1283.50	347.08 ± 433.24	542.22 ± 662.81	***
LF/HF (ms ²)	2.8 ± 2.6 [62]	1.65 ± 1.60	2.53 ± 1.72	3.28 ± 2.95	2.84 ± 2.77	*
<i>LF_{nu}</i>	0.46 ± 0.20 [24]	0.51 ± 0.23	0.63 ± 0.20	0.68 ± 0.16	0.64 ± 0.19	–
SD1 (ms)	20.8 ± 9.8 [44]	55.51 ± 36.72	42.88 ± 31.45	34.82 ± 26.57	27.46 ± 23.73	**
SD2 (ms)	63.3 ± 16.4 [44]	120.09 ± 70.31	87.02 ± 59.68	74.47 ± 42.02	66.14 ± 30.43	**
SD1/SD2 (ms)	0.35 ± 0.08 [63]	0.47 ± 0.19	0.47 ± 0.15	0.43 ± 0.20	0.37 ± 0.16	*
SampEn (arb. units)	1.34 ± 0.54 [41]	1.21 ± 0.61	1.39 ± 0.56	1.04 ± 0.39	1.14 ± 0.46	**
α1 (arb. units)	1.02 ± 0.18 [44]	1.02 ± 0.32	1.05 ± 0.26	1.16 ± 0.30	1.27 ± 0.23	**

Signif. codes: *** *p* < 0.0001 ** *p* < 0.001, * *p* < 0.01, • *p* < 0.05

in the Table 2 are from the ANOVA test on the TSST phases. In the Tukey multiple comparison test, the parameters with significant differences between all the stress and non-stress phases of TSST are marked in the Table 2 as bold, and they are AVNN, rMSSD, SDNN and pNN20.

Results of the pair-wise analysis of Tukey test on the HRV metrics analysis showed that AVNN, SDNN, rMSSD and pNN20 metrics had significant differences between all combinations of stress and non-stress phases, i.e reading-presentation, silence-presentation, silence-counting and reading-counting, even in 50 s windows analysis. The pNN50 was the time domain metric with no significant results between stress and non-stress phases, with no significant differences for reading-presentation, reading-counting and presentation-counting.

For cases where the *p*-value by Tukey test showed significant differences between stress and non-stress TSST phases (AVNN, SDNN, rMSSD and pNN20) were represented in the Fig. 5 with respective confidence intervals.

Fig. 5 contains confidence intervals for the difference in the means for all pair-wise from the four TSST phases. In this graph, confidence intervals that include 0 indicate HRV metrics that are not significantly different (*p* < 0.5). AVNN showed significant differences for all pair-wises: silence-reading, silence-presentation, silence-counting, reading-presentation, reading-counting and presentation-counting. SDNN and rMSSD did not show significant differences between presentation-counting and silence-reading. pNN20 just did not show significant differences between presentation-counting.

Fig. 6 represents the significant *p*-values for each pair-wise obtained for the 50 s window length AVNN analysis. For the AVNN the results between stress and non-stress were highly statistically significant with *p* < 0.0001. Even among the phases from baseline: silence-reading (*p* < 0.0001) or from the experiment: presentation-counting (*p* < 0.05) this metric showed significant differences represented in the Fig. 6.

In the frequency domain metrics, the HF showed better results than the other metrics in this domain. HF had significant differences between TSST phases, however between reading and counting, stress Vs non-stress, the HF metric did not show a statistical difference. The metrics LF and LF/HF showed significant differences between the TSST for the ANOVA test, but did not show significant results between the stress and non-stress phases for all situations. The LF did not show significant results between silence-counting and reading-counting, and the LF/HF not showed statistical differences between reading-counting and reading-presentation. From nonlinear analysis, the SD1 and SD2 are the metrics that showed significant differences between silence and other TSST phases,

however, it did not present statistical differences between stress and non-stress phases, between reading-presentation and reading-counting did not show significant results. Other non-linear metrics showed statistical differences between TSST phases, however, without significant *p*-values for some combinations between stress and non-stress phases.

The fine-grained analysis allows the representation of the HRV metrics variation during the entire TSST, and stress level assessment almost in a continuous way, for the AVNN metric in the Fig. 7. The reading task belongs to the baseline (phase without stress) and allows to assess the effect of the speaking activity. The free speech and arithmetic counting activity are demanding cognitive tasks with higher difficult level and consequently induce the mental stress. The differences induced by the TSST procedure and assessed by the AVNN are clearly visible in the Fig. 7.

4. Discussion

In this section the results for HRV metrics obtained with short-time windows of 50 s length are discussed, since for higher length of windows analysis all the metrics show a significant differences between TSST phases, and the main objective of this work was to evaluated the capability of HRV metrics to provide a fine-grained stress assessment and select the descriptors which could be used in short-time windows.

The differences verified from silence to speaking tasks (reading, presentation and counting) reflect the effect of speaking activity. Respiratory changes produced by speech markedly alter variability of HRV without necessarily involving changes in autonomic activation. Increased respiration induce differences between HR in silence and talking conditions [28,64]. The respiration effect is the main contribution for HRV differences between silence and reading, both in the baseline phase without stress. Nevertheless, HRV metrics reveal significant differences between other TSST stages, that confirms the main influence of ANS on the changes in the heart rate variability metrics.

The AVNN, SDNN and pNN50 were expected to be significantly reduced during the psychological stressors activation in the experiment phase (speech, mental arithmetic) compared to the baseline [22,24,25,52]. The results obtained in the present work confirmed the reduction of these metrics with stress condition. The AVNN is an indicator of the ratio of the cardiac sympathetic to parasympathetic tones (sympatho-vagal balance). The trend of AVNN proves its efficacy in distinguishing mental stress. AVNN metric showed that it is the most reliable metric to recognize mental stress across the four phases of the TSST procedure [52] for each size of window analysis in this study. The elevated AVNN during the stress

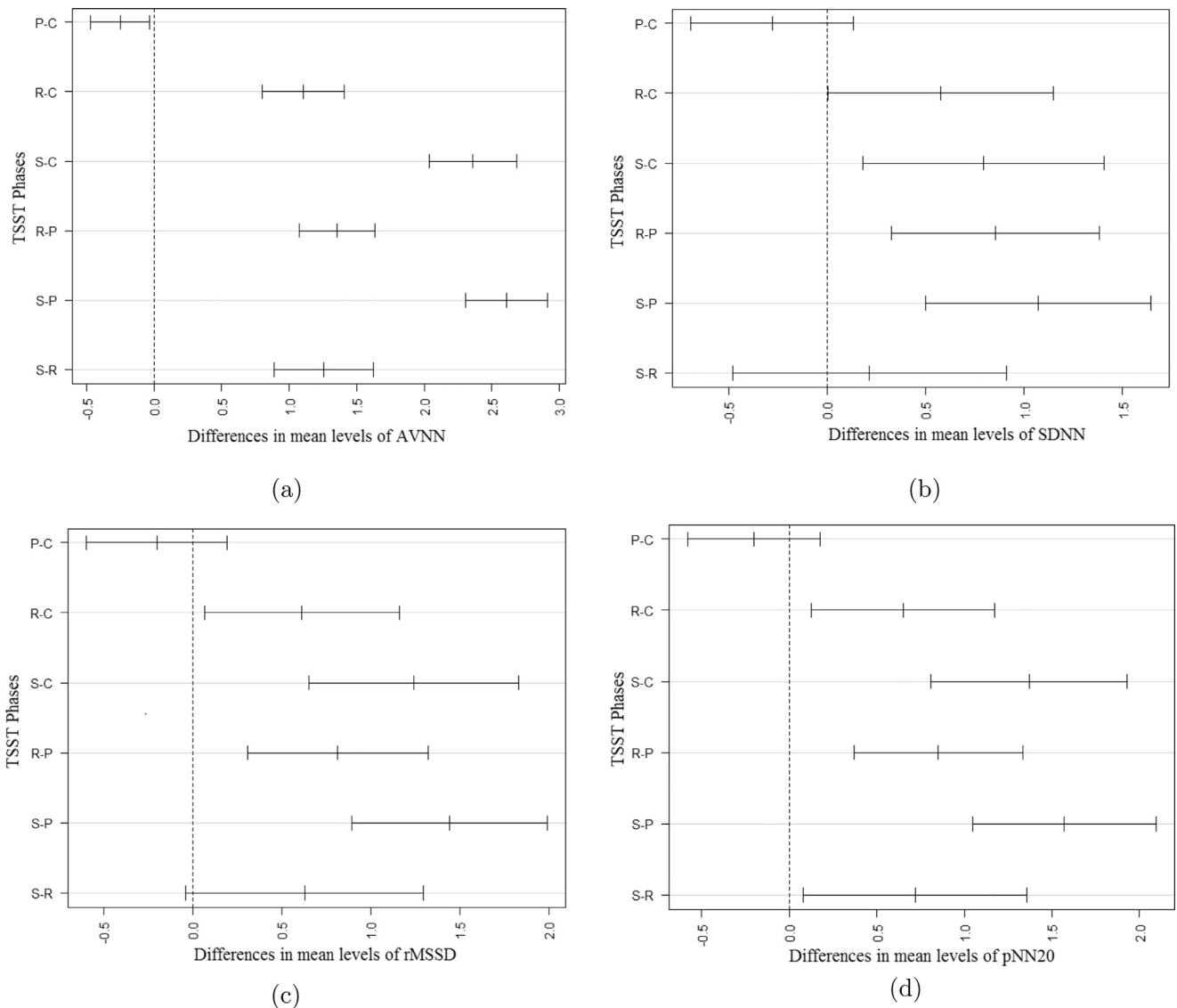


Fig. 5. Graphical display of 95% individual confidence levels of pair-wise comparisons from Tukey's HRV metrics for the TSST phases (S- silence, R - reading, P - presentation, C - counting). Four HRV metrics that showed a significance differences between the stressful and non-stress phase using 50 s window length are represented: (a) confidence intervals for AVNN; (b) SDNN; (c) rMSSD; and, (d) pNN20 metrics.

phase indicate the under-arousal of the parasympathetic system and/or over-arousal of sympathetic nervous system [65]. The significant decrease in SDNN during the experiment phase reflects a decreased variability in HR during the presence of mental stress [24], although the SDNN parameter did not show significant differences between baseline phases (silence-reading). The SDNN reflects all the cyclic components responsible for variability in the period of recording. The significant decrease in SDNN with stress condition reflects a decreased total HRV in the presence of mental stress [24]. SDNN represents the global autonomic function of the heart, joint sympathetic and parasympathetic modulation of heart rate, whereas rMSSD and pNN50 are thought to represent parasympathetic modulation of heart rate [66–68]. The rMSSD and pNN20 metrics correspond to short-term HRV changes, reflect alterations in autonomic nervous system and have been considered good metrics for stress assessment [16], however the results obtained for each metric show significant pairwise differences between the TSST phases, with absence of significance only between presentation and counting. Recent studies concluded that pNN20 is highly negatively correlated to stress and may be more sensi-

tive than the traditional pNN50 to figure out aspects in physiological response [59]. The pNN50 did not show significant differences between several phases (reading-presentation; presentation-counting; reading-counting), even among stress and non-stress condition, and confirms higher correlation of pNN20 with stress compared to pNN50.

The spectral analysis verified that stress induced a decrease in the HF values associated with a significant reduction in parasympathetic cardiac control over the heart [29,69]. The results obtained for HF parameter showed significant pair-wise differences between the TSST tasks in short-time analysis of 50 s length window, however, were not significant in reading-counting. The decrease in high frequency power, and an increase in the LF/HF ratio with the stress phases [29], were verified in this study. This tendency indicates higher sympathetic activity with a mental task compared with the baseline phase [25], however this parameter did not show a significant difference between the stress and non-stress (reading-presentation and reading-counting). The HF power of RR-intervals decreased as evidence of withdrawal of vagal activity, and the LF/HF increases showing the most influence of sym-

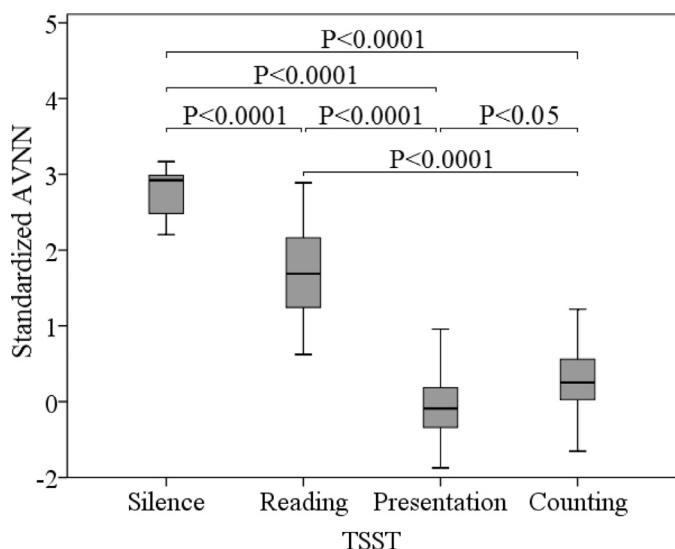


Fig. 6. Boxplot of standardized AVNN for all subjects.

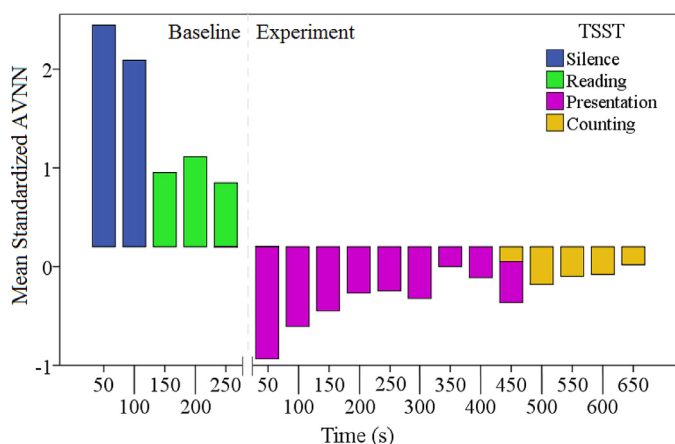


Fig. 7. Plot of mean standardized AVNN for all subjects in 50 s of window length analysis during the TSST.

pathetic system during the stress condition. Disagreement exists in respect of physiological interpretation of LF and LF_{nu} component [15]. LF decreases with stress condition but no show statistical significant difference between the stress and non-stress phases, according to other studies [22,26,38].

Nonlinear metrics have been less applied in stress assessment, however, SD1 and α_1 are metrics that reflect short-term variability [70]. SD1 parameter showed significantly higher values in non-stress condition than in stress condition, reflecting a significant inhibition of the parasympathetic activity [71]. α_1 was correlated with sympatho-vagal activity yielded [72], however, showed significant changes with stress levels in this study [72]. α_1 values higher and far from 1.0 represent decreased parasympathetic and increased sympathetic modulation of the heart [73]. The changes in the LF/HF ratio have been interpreted as reflecting sympathetic modulation [74], based on the mathematical argument, α_1 is related to the LF/HF ratio [75], and these parameters reflect the sympatho-vagal balance. SD2 show a decrease with stress condition according to the other studies [23,76,77], and a significant difference between baseline and experiment phase. SD2 reflects the continuous long-term variability of the RR intervals however, remains with uncertain physiological meaning and interpretation [77]. Increasing stress led to a significant decrease in SD1/SD2 value [77]. SD1/SD2 ratio is interpreted as a measure of the bal-

ance between short and long-term HRV. The short RR interval corresponding to low complexity means low SampEn values. The metric SampEn is correlated to HF metric by a quadratic function [78]. The increase of the stress condition decrease the values of SampEn [77]. A low SampEn designates a high regularity and may be linked to high levels of stress, an increased SampEn corresponds to an increase in randomness in the data, a relaxed state (baseline) [79]. Some limitations of the sample entropy estimations were studied and theoretical basis on which this metric is constructed has been questioned [80]. The referred study advise that the use of the sample entropy in the interbeat intervals measurements must be carefully applied whereas the direct estimation from the observed signal would combine dynamical and observational uncertainties [80].

The results obtained in this work show the higher performance of time domain parameters for stress assessment [14,21,24]. In the global analysis, the AVNN parameter shows a significant difference between the four TSST phases, for all windows length in analysis. According to the results in this paper, the sub-set of metrics that allow to distinguish stress from non-stress condition using a fine-grained analysis can be chosen as follows: AVNN, rMSSD, SDNN and pNN20.

Study limitations

The limit of 50 s for the windows length analysis was imposed by the accurate estimation of the LF metric. However, in case of time domain metrics it could be possible to use smaller windows, and obtain stress levels at more granular time intervals.

The TSST represents a method that induces stress in blocks of time periods, during the experiment stage the subject is under the stress situation. In the real life, the subjects live with several situations with stressors agents that trigger the stress but it represents a point in the time, and the subject recovery after that [81]. The conclusions of the present work could be applied to the situation of a mental effort such as an exam or job interview, that represents a long period under the stress, but more studies will be necessary to understand the response of HRV metrics in the short fright situations.

This study was limited to four specific activities and a relatively short recording time. The next step will be to design a test that enables more tasks and comparison with other measurements such as the galvanic skin response. The relation and time response between a stimulus and the physiological effect of a stressor could be studied with HRV metrics.

One limitation of this study was the reduced age span of participants examined and some possible relations between stressors and outcomes may not have been clearly shown. However, in spite of the reduced cohort size, it is enough to draw conclusions about the significant differences between moments were evident and important relations between stress and HRV metrics emerged.

5. Conclusion

The HRV metrics were used in short term analysis to assess the stress condition and different lengths were tested: 220 s, 150 s, 100 s and 50 s. This work demonstrates that the AVNN metric, using 50 s of windows length analysis, allows a fine-grained analysis of stress effect as an index of psychological stress and provide a better insight into the reaction of the autonomic nervous system to stress. This work represents an extensive characterization of HRV that enables the selection of the metrics sub-set which could be used in short-time windows to assess the physiological correlation of stress. Globally, time domain methods showed to be more robust in detecting stress than frequency domain or nonlinear approaches. The overall result is that AVNN, RMSSD, SDNN and pNN20 are the most discriminating HRV metrics to differentiate between stress and non-stress with a 50 s window.

As a future work, the authors propose the use of machine learning techniques and a larger HRV dataset from a big sample of subjects, in order to develop an automatic method to classify the stress level using the basal subject condition in real context, in order to monitor the stress lived by the subjects during their lives. Another challenge for the future is the test of small windows length with time domain metrics to assess the stress levels.

Acknowledgments

Research supported by grants PTDC/EEAELC/121018/2010 (VOCE) and in the scope of R&D Unit UID/EEA/50008/2013, financed by the applicable financial framework (FCT/MEC through national funds and when applicable co-funded by FEDER PT2020 partnership agreement).

References

- [1] J. Dimsdale, Psychological stress and cardiovascular disease, *J. Am. Coll. Cardiol.* 51 (13) (2008) 1237–1246, doi:10.1016/j.jacc.2007.12.024.
- [2] S. Cohen, D. Janicki-Deverts, W.J. Doyle, G.E. Miller, E. Frank, B.S. Rabin, R.B. Turner, Chronic stress, glucocorticoid receptor resistance, inflammation, and disease risk, *Proc. Natl. Acad. Sci. U.S.A.* 109 (16) (2012) 5995–5999, doi:10.1073/pnas.1118355109.
- [3] B.S. Oken, I. Chamine, W. Wakeland, A systems approach to stress, stressors and resilience in humans, *Behav. Brain Res.* 282 (2015) 144–154, doi:10.1016/j.bbr.2014.12.047.
- [4] J. Choi, G.O. Ricardo, Using heart rate monitors to detect mental stress, *Proceedings - 2009 6th International Workshop on Wearable and Implantable Body Sensor Networks, BSN 2009* (2009) 219–223, doi:10.1109/BSN.2009.13.
- [5] Y.J. Zaanstra, D.W. Johnston, Cardiovascular reactivity in real life settings: measurement, mechanisms and meaning, *Biol. Psychol.* 86 (2) (2011) 98–105, doi:10.1016/j.biopsycho.2011.07.011.
- [6] J. Cunha, B. Cunha, A. Pereira, W. Xavier, N. Ferreira, L. Meireles, Vital-jacket: a wearable wireless vital signs monitor for patients' mobility in cardiology and sports, in: *Pervasive Computing Technologies for Healthcare (PervasiveHealth)*, 2010 4th International Conference, 2010, pp. 9–10, doi:10.4108/ICST.PERVASIVEHEALTH2010.8991.
- [7] J. Gao, B.M. Gurbaxani, J. Hu, K.J. Heilman, V.A. Emanuele, G.F. Lewis, M. Davila, E.R. Unger, J.M.S. Lin, Multiscale analysis of heart rate variability in non-stationary environments, *Front Physiol* 4 MAY (May) (2013) 1–8, doi:10.3389/fphys.2013.00119.
- [8] G. Valenza, R.G. Garcia, L. Citi, E.P. Scilingo, C.A. Tomaz, R. Barbieri, Nonlinear digital signal processing in mental health: characterization of major depression using instantaneous entropy measures of heartbeat dynamics, *Front Physiol.* 6 (MAR) (2015) 1–8, doi:10.3389/fphys.2015.00074.
- [9] G. Berntson, J.T. Cacioppo, P. Binkley, B. Uchino, K. Quigley, A. Fieldstone, Autonomic cardiac control: III. Psychological stress and cardiac response in autonomic space as revealed by pharmacological blockades, *Psychophysiology* 31 (1994) 599–608.
- [10] A. De Santos Sierra, C. Sánchez Ávila, J. Guerra Casanova, G. Bailador Del Pozo, A stress-detection system based on physiological signals and fuzzy logic, *IEEE Trans. Ind. Electron.* 58 (10) (2011) 4857–4865, doi:10.1109/TIE.2010.2103538.
- [11] M. Rahman, D.H. Epstein, K.L. Preston, M. Jobs, J.G. Beck, S. Kedia, K.D. Ward, M. Al'Absi, S. Kumar, R. Bari, A.A. Ali, M. Sharmin, A. Raji, K. Hovsepian, S.M. Hossain, E. Ertin, A. Kennedy, Are we there yet? Feasibility of Continuous stress assessment via wireless physiological sensors, in: *Proceedings of the 5th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics - BCB '14*, 2014, pp. 479–488, doi:10.1145/2649387.2649433.
- [12] P. Jonsson, M. Wallergard, K. Osterberg, A.M. Hansen, G. Johansson, B. Karlson, Cardiovascular and cortisol reactivity and habituation to a virtual reality version of the trier social stress test: a pilot study, *Psychoneuroendocrinology* 35 (9) (2010) 1397–1403, doi:10.1016/j.psyneuen.2010.04.003.
- [13] S. Shiffman, A.A. Stone, M.R. Hufford, Ecological momentary assessment, *Annu. Rev. Clin. Psychol.* 4 (2008) 1–32, doi:10.1146/annurev.clinpsy.3.022806.091415.
- [14] E.B. Schroeder, E.A. Whitsel, G.W. Evans, R.J. Prineas, L.E. Chambless, G. Heiss, Repeatability of heart rate variability measures, *J. Electrocardiol.* 37 (3) (2004) 163–172, doi:10.1016/j.jelectrocard.2004.04.004.
- [15] Task Force of The European Society of Cardiology and The North American Electrophysiology Society of Pacing and Electrophysiology, Guidelines heart rate variability, *Eur. Heart J.* 17 (1996) 354–381.
- [16] J. Sztajzel, Heart rate variability: a noninvasive electrocardiographic method to measure the autonomic nervous system, *Swiss Med. Wkly* 134 (35–36) (2004) 514–522.
- [17] X. Wang, X. Ding, S. Su, Z. Li, H. Riese, J.F. Thayer, F. Treiber, H. Snieder, Genetic influences on heart rate variability at rest and during stress, *Psychophysiology* 46 (3) (2009) 458–465, doi:10.1111/j.1469-8986.2009.00793.x.
- [18] M. De Rivecourt, M.N. Kuperus, W.J. Post, L.J.M. Mulder, Cardiovascular and eye activity measures as indices for momentary changes in mental effort during simulated flight, *Ergonomics* 51 (9) (2008) 1295–1319, doi:10.1080/00140130802120267.
- [19] L. Salahuddin, J. Cho, M.G. Jeong, D. Kim, Ultra short term analysis of heart rate variability for monitoring mental stress in mobile settings, in: *Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society*, 2007, 2007, pp. 4656–4659, doi:10.1109/IEMBS.2007.4353378.
- [20] G.D. Clifford, *Signal Processing Methods for Heart Rate Variability*, Doctoral thesis, University of Oxford, 2002.
- [21] A. Arza, J.M. Garzón, A. Hemando, J. Aguiló, R. Bailon, Towards an objective measurement of emotional stress: Preliminary analysis based on heart rate variability, in: *37th Annual International Conference of IEEE Engineering in Medicine and Biology Society*, 2015, pp. 3331–3334.
- [22] D.A. Dimitriev, A.D. Dimitriev, Y.D. Karpenko, E.V. Saperova, Influence of examination stress and psychoemotional characteristics on the blood pressure and heart rate regulation in female students, *Hum. Physiol.* 34 (5) (2008) 617–624, doi:10.1134/S0362119708050101.
- [23] P. Melillo, M. Bracale, L. Pecchia, Nonlinear heart rate variability features for real-life stress detection. case study: students under stress due to university examination, *Biomed. Eng. Online* 10 (1) (2011) 96, doi:10.1186/1475-925X-10-96.
- [24] E. Tharion, S. Parthasarathy, N. Neelakantan, Short-term heart rate variability measures in students during examinations, *Natl. Med. J. India* 22 (2) (2009) 63–66.
- [25] J. Taelman, S. Vandeput, A. Spaepen, S.V. Huffel, Influence of mental stress on heart rate and heart rate variability, in: *International Federation for Medical and Biological Engineering Proceedings* 2008, 29, 2008, pp. 1366–1369, doi:10.1007/978-3-540-89208-3_324.
- [26] D. Lucini, G. Norbiato, M. Clerici, M. Pagani, Hemodynamic and autonomic adjustments to real life stress conditions in humans, *Hypertension* 39 (1) (2002) 184–188, doi:10.1161/hy0102.100784.
- [27] J. Hellhammer, M. Schubert, The physiological response to trier social stress test relates to subjective measures of stress during but not before or after the test, *Psychoneuroendocrinology* 37 (1) (2012) 119–124, doi:10.1016/j.psyneuen.2011.05.012.
- [28] C. Schubert, M. Lambertz, R. Nelesen, W. Bardwell, J.-B. Choi, J. Dimsdale, A. Effects of stress on heart rate complexity: a comparison between short-term and chronic stress, *Biol. Psychol.* 80 (3) (2009) 325–332, doi:10.1016/j.biopsycho.2008.11.005.
- [29] G.G. Berntson, J.T. Cacioppo, Heart rate variability: stress and psychiatric conditions, in: *Dynamic Electrocardiography*, 2007, pp. 57–64, doi:10.1002/9780470987483.ch7.
- [30] J.N. Jeffrey, J. Goldberger, *Practical signal and image processing in clinical cardiology*, 2010.
- [31] E. von Borell, J. Langbein, G. Després, S. Hansen, C. Lettieri, J. Marchant-Forde, R. Marchant-Forde, M. Minero, E. Mohr, A. Prunier, D. Valance, I. Veissier, Heart rate variability as a measure of autonomic regulation of cardiac activity for assessing stress and welfare in farm animals - a review, *Physiol. Behav.* 92 (3) (2007) 293–316, doi:10.1016/j.physbeh.2007.01.007.
- [32] A. Aguiar, M. Kaiseler, H. Meinedo, T.E. Abrudan, P.R. Almeida, Speech stress assessment using physiological and psychological measures, in: *Proceedings of the 2013 ACM Conference on Pervasive and Ubiquitous Computing Adjunct Publication - UbiComp*, 2013, pp. 921–930, doi:10.1145/2494091.2497346.
- [33] A.L. S. H. Goldberger, L.A.N. Amaral, L. Glass, J.M. Hausdorff, Pch Ivanov, R.G. Mark, J.E. Mietus, G.B. Moody, C.K. Peng, Physiobank, physiotoolkit, and physionet components of a new research resource for complex physiologic signals, *Circulation* 101 (2000) e215–e220.
- [34] C.D. Spielberger, *State-trait anxiety inventory*, *Corsini Encyclopedia of Psychology* 1, 2010.
- [35] P. Gomes, M. Kaiseler, C. Queiros, M. Oliveira, B. Lopes, M. Coimbra, Vital analysis: annotating sensed physiological signals with the stress levels of first responders in action, in: *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS*, 2012, pp. 6695–6698, doi:10.1109/EMBS.2012.6347530.
- [36] J. Pallauf, P. Gomes, S. Brás, J.P.S. Cunha, M. Coimbra, Associating ECG features with firefighter's activities, in: *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS*, 2011, pp. 6009–6012, doi:10.1109/IEMBS.2011.6091485.
- [37] R. Shaffer, M. McCraty, C.L. Zerr, D.V.A. Medical, A healthy heart is not a metronome: an integrative review of the heart's anatomy and heart rate variability, *Front Psychol.* 5 (September) (2014) 1–19, doi:10.3389/fpsyg.2014.01040.
- [38] R.L. Burr, Interpretation of normalized spectral heart rate variability indices in sleep research: a critical review, *Sleep* 30 (7) (2007) 913–919.
- [39] J.M. Yentes, N. Hunt, K.K. Schmid, J.P. Kaipust, D. McGrath, N. Stergiou, The appropriate use of approximate entropy and sample entropy with short data sets, *Ann. Biomed. Eng.* 41 (2) (2013) 349–365, doi:10.1007/s10439-012-0668-3.
- [40] S. Ahmad, M. Bolic, H. Dajani, V. Groza, I. Batkin, S. Rajan, Measurement of heart rate variability using an oscillometric blood pressure monitor, *IEEE Trans. Instrum. Meas.* 59 (10) (2010) 2575–2590, doi:10.1109/TIM.2010.2057571.
- [41] R. Cabiddu, R. Trimer, A. Borghi-Silva, M. Migliorini, R.G. Mendes, A.D. Oliveira Jr., F.S.M. Costa, A.M. Bianchi, Are complexity metrics reliable in assessing HRV control in obese patients during sleep? *PLoS ONE* 10 (4) (2015) e0124458, doi:10.1371/journal.pone.0124458.
- [42] M. Weippert, M. Behrens, A. Rieger, K. Behrens, Sample entropy and traditional measures of heart rate dynamics reveal different modes of cardiovascular control during low intensity exercise, *Entropy* 16 (11) (2014) 5698–5711, doi:10.3390/e16115698.

- [43] V.D.a. Corino, M. Matteucci, L. Cravello, E. Ferrari, A.a. Ferrari, L.T. Mainardi, Long-term heart rate variability as a predictor of patient age, *Comput. Methods Programs Biomed.* 82 (3) (2006) 248–257, doi:[10.1016/j.cmpb.2006.04.005](https://doi.org/10.1016/j.cmpb.2006.04.005).
- [44] A. Voss, S. Schulz, R. Schroeder, M. Baumert, P. Caminal, Methods derived from nonlinear dynamics for analysing heart rate variability, *Philos. Trans. A Math. Phys. Eng. Sci.* 367 (1887) (2009) 277–296, doi:[10.1098/rsta.2008.0232](https://doi.org/10.1098/rsta.2008.0232).
- [45] E. Sarabia-cachadi, B. De, L.R. Alvarez, R. Juan, Heart rate variability non linear dynamics in intense exercise, in: *Computing in Cardiology*, 2012, pp. 177–180.
- [46] G. Tanev, D.B. Saadi, K. Hoppe, H.B. Sorensen, Classification of acute stress using linear and non-linear heart rate variability analysis derived from sternal ECG, in: 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society to be Held in August 2014, 2014, pp. 3386–3389, doi:[10.1109/EMBC.2014.6944349](https://doi.org/10.1109/EMBC.2014.6944349).
- [47] M.A. Peltola, Role of editing of R-R intervals in the analysis of heart rate variability, *Front Physiol.* 3 (2012) 1–10, doi:[10.3389/fphys.2012.00148](https://doi.org/10.3389/fphys.2012.00148).
- [48] A.E. Aubert, D. Ramaekers, F. Beckers, R. Breem, C. Deneef, F. Van de Werf, H. Ector, The analysis of heart rate variability in unrestrained rats. validation of method and results., *Comput. Methods Programs Biomed.* 60 (3) (1999) 197–213.
- [49] G. Moody, Spectral analysis of heart rate without resampling, in: *Proceedings - Computers in Cardiology*, 1993, pp. 7–10.
- [50] B.S. Saini, D. Singh, M. Uddin, V. Kumar, Improved power spectrum estimation for RR-Interval time series, *Int. J. Electr. Comput. Energet. Electron. Commun. Eng.* 2 (10) (2008) 154–158.
- [51] J. Ramshur, Design, Evaluation, and Application of Heart Rate Variability Analysis Software (Hrvas), Ph.D. thesis, University of Memphis, 2010.
- [52] F. Sun, C. Kuo, H. Cheng, S. Buthpitiya, P. Collins, M. Griss, Activity-aware mental stress detection using physiological sensors, in: M. Gris, G. Yang (Eds.), *Mobile Computing, Applications, and Services*, Springer Berlin Heidelberg, 2012, pp. 211–230.
- [53] J. McNames, M. Aboy, Reliability and accuracy of heart rate variability metrics versus ECG segment duration, *Med. Biol. Eng. Comput.* 44 (9) (2006) 747–756, doi:[10.1007/s11517-006-0097-2](https://doi.org/10.1007/s11517-006-0097-2).
- [54] P. Laguna, G.B. Moody, R.G. Mark, Power spectral density of unevenly sampled data by least-square analysis: performance and application to heart rate signals *IEEE trans, Biomed. Eng.* 45 (6) (1998) 698–715.
- [55] J.E. Molina, Heart rate variability standards, *Circulation* 98 (15) (1998) 1587–1588, doi:[10.1161/01.CIR.98.15.1587](https://doi.org/10.1161/01.CIR.98.15.1587).
- [56] G.D. Clifford, ECG statistics, noise, artifacts, and missing data, in: F.A.G.D. Clifford, P.E. McSharry (Eds.), *Advanced Methods for ECG Analysis*, Artech House, London, 2006.
- [57] L.M. Sullivan, Repeated measures, *Circulation* 117 (9) (2008) 1238–1243, doi:[10.1161/CIRCULATIONAHA.107.654350](https://doi.org/10.1161/CIRCULATIONAHA.107.654350).
- [58] N. Crichton, Tukey multiple comparison test, *J. Clin. Nurs.* 8 (1995) 299–304.
- [59] M. Trimmel, Relationship of heart rate variability (HRV) parameters including pnnxx with the subjective experience of stress, depression, well-being, and every-day trait moods (TRIM-T): A Pilot study, *Ergonomics Open* J. 8 (2015) 32–37.
- [60] T.V. Sousa, V. Viveiros, M.V. Chai, F.L. Vicente, G. Jesus, M.J. Carnot, A.C. Gordo, P.L. Ferreira, Reliability and validity of the portuguese version of the generalized anxiety disorder (GAD-7) scale., *Health Qual Life Outcomes* 13 (1) (2015) 50, doi:[10.1186/s12955-015-0244-2](https://doi.org/10.1186/s12955-015-0244-2).
- [61] H. Bonnemeier, U.K.H. Wiegand, A. Brandes, N. Kluge, H.a. Katus, G. Richardt, J. Potratz, Circadian profile of cardiac autonomic nervous modulation in healthy subjects: differing effects of aging and gender on heart rate variability, *J. Cardiovasc. Electrophysiol.* 14 (8) (2003) 791–799, doi:[10.1046/j.1540-8167.2003.03078.x](https://doi.org/10.1046/j.1540-8167.2003.03078.x).
- [62] D. Nunan, G.R.H. Sandercock, D.A. Brodie, A quantitative systematic review of normal values for short-term heart rate variability in healthy adults, *PACE - Pacing Clin. Electrophysiol.* 33 (11) (2010) 1407–1417, doi:[10.1111/j.1540-8159.2010.02841.x](https://doi.org/10.1111/j.1540-8159.2010.02841.x).
- [63] L.C.M. Vanderlei, C.M. Pastre, I.F. Freitas, M.F.D. Godoy, Geometric indexes of heart rate variability in obese and eutrophic children, *Arq. Bras. Cardiol.* 95 (1) (2010) 35–40.
- [64] V.A. MacIntyre, P.D. MacIntyre, G. Carre, Heart rate variability as a predictor of speaking anxiety, *Commun. Res. Reports* 27 (4) (2010) 286–297, doi:[10.1080/08824096.2010.496323](https://doi.org/10.1080/08824096.2010.496323).
- [65] A. Kushki, E. Drumm, M. Pla Mobarak, N. Tanel, A. Dupuis, T. Chau, E. Anagnostou, Investigating the autonomic nervous system response to anxiety in children with autism spectrum disorders, *PLoS ONE* 8 (4) (2013) 2–9, doi:[10.1371/journal.pone.0059730](https://doi.org/10.1371/journal.pone.0059730).
- [66] M.R. Carnethon, R.J. Prineas, M. Temprosa, Z.M. Zhang, G. Uwaifo, M.E. Molitch, The association among autonomic nervous system function, incident diabetes, and intervention arm in the diabetes prevention program, *Diabetes Care* 29 (4) (2006) 914–919, doi:[10.2337/diacare.29.04.06.dc05-1729](https://doi.org/10.2337/diacare.29.04.06.dc05-1729).
- [67] M. Todo, K. Yamauchi, T. Arahira, Effects of massage therapy intervention on autonomic nervous system promotion in integrated circuit design company employees, in: *IFMBE Proceedings*, 43, 2014, pp. 865–867, doi:[10.1007/978-3-319-02913-9](https://doi.org/10.1007/978-3-319-02913-9).
- [68] G.S.C. Barreto, F.M. Vanderlei, L.C.M. Vanderlei, Á.J.M. Leite, Impact of malnutrition on cardiac autonomic modulation in children, *J. Pediatr* (xx) (2016), doi:[10.1016/j.jpeds.2016.03.005](https://doi.org/10.1016/j.jpeds.2016.03.005).
- [69] J. Taelman, S. Vandeput, I. Gligorijević, A. Spaepen, S. Van Huffel, Time-frequency heart rate variability characteristics of young adults during physical, mental and combined stress in laboratory environment, in: *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, EMBS, 96, 2011, pp. 1973–1976, doi:[10.1109/IEMBS.2011.6090556](https://doi.org/10.1109/IEMBS.2011.6090556).
- [70] P.K. Stein, A. Reddy, Non-linear heart rate variability and risk stratification in cardiovascular disease, *Indian Pacing Electrophysiol. J.* 5 (3) (2005) 210–220.
- [71] J.C. Cervantes Blázquez, G. Rodas Font, L. Capdevila Orts, Heart-rate variability and precompetitive anxiety in swimmers, *Psicothema* 21 (4) (2009) 531–536.
- [72] J. Rojo-Alvarez, A. Sanchez-Sanchez, O. Barquero-Perez, R. Goya-Esteban, E. Everss, I. Mora-Jimenez, A. Garcia-Alberola, Analysis of physiological meaning of detrended fluctuation analysis in heart rate variability using a lumped parameter model, in: *Computers in Cardiology*, 2007, pp. 25–28, doi:[10.1109/CIC.2007.4745412](https://doi.org/10.1109/CIC.2007.4745412).
- [73] A.C. de Souza, J.R. Cisternas, L.C. de Abreu, A.L. Roque, C.B. Monteiro, F. Adami, L.C.M. Vanderlei, F.H. Sousa, L.L. Ferreira, V.E. Valenti, Fractal correlation property of heart rate variability in response to the postural change maneuver in healthy women, *Int. Arch. Med.* 7 (2014) 25, doi:[10.1186/1755-7682-7-25](https://doi.org/10.1186/1755-7682-7-25).
- [74] K. Petrowski, U. Herold, P. Joraschky, M. Mück-Weymann, M. Siepmann, The effects of psychosocial stress on heart rate variability in panic disorder, *German J. Psychiatry* 13 (2) (2010) 66–73.
- [75] S.O. Satoshi IwaseJunichiro Hayano, Heart rate variability and sympathetic nerve activity, in: *Clinical Assessment of the Autonomic Nervous System*, Springer, 2016, p. 317.
- [76] R. Orsila, M. Virtanen, M. Savinainen, Perceived mental stress and reactions in heart rate variability a pilot study among employees of an electronics company, *Int. J. Occup. Saf. Ergonom.* 14 (3) (2008) 275–283.
- [77] D.A. Dimitriev, E.V. Saperova, A.D. Dimitriev, State anxiety and nonlinear dynamics of heart rate variability in students, *PLoS ONE* 11 (1) (2016) 1–22, doi:[10.1371/journal.pone.0146131](https://doi.org/10.1371/journal.pone.0146131).
- [78] M.M. Platasa, V. Gal, Reflection of heart rate regulation on linear and nonlinear heart rate variability measures, *Physiol. Meas.* 27 (November 2016) (2006) 145–154, doi:[10.1088/0967-3334/27/2/005](https://doi.org/10.1088/0967-3334/27/2/005).
- [79] T. Chanwimalueang, L. Aufegger, W. Von Rosenberg, D.P. Mandic, Modelling stress in public speaking: evolution of stress levels during conference presentations, in: *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*, 2016– May, 2016, pp. 814–818, doi:[10.1109/ICASSP.2016.7471788](https://doi.org/10.1109/ICASSP.2016.7471788).
- [80] D. Darmon, Specific differential entropy rate estimation for continuous-valued time series, *Entropy* 18 (5) (2016) 190, doi:[10.3390/e18050190](https://doi.org/10.3390/e18050190).
- [81] J.G.P. Rodrigues, M. Kaiseler, A. Aguiar, J. Barros, J.P.S. Cunha, A mobile sensing approach to stress detection and memory activation for public bus drivers, *Intell. Transport. Syst. IEEE Trans. PP* (99) (2015) 1–10.